

Project Management

2. PMI Process Groups & Knowledge Areas

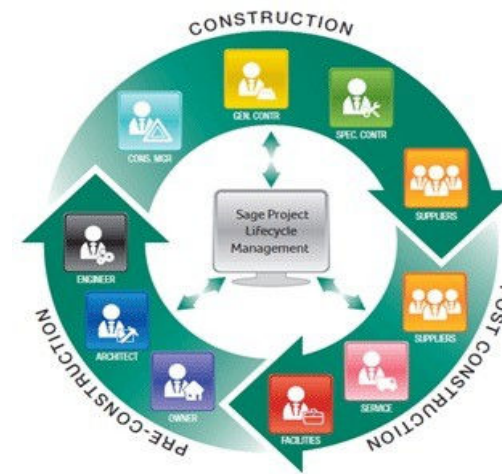
Contents

- Project life cycle
- Approaches in project management
- Project phases and decision points
- Decision making tools:
Cynefine Framework and
Stacey Matrix
- 5 groups of project management processes.
- 10 areas in project management.
- Qualities of a project manager.

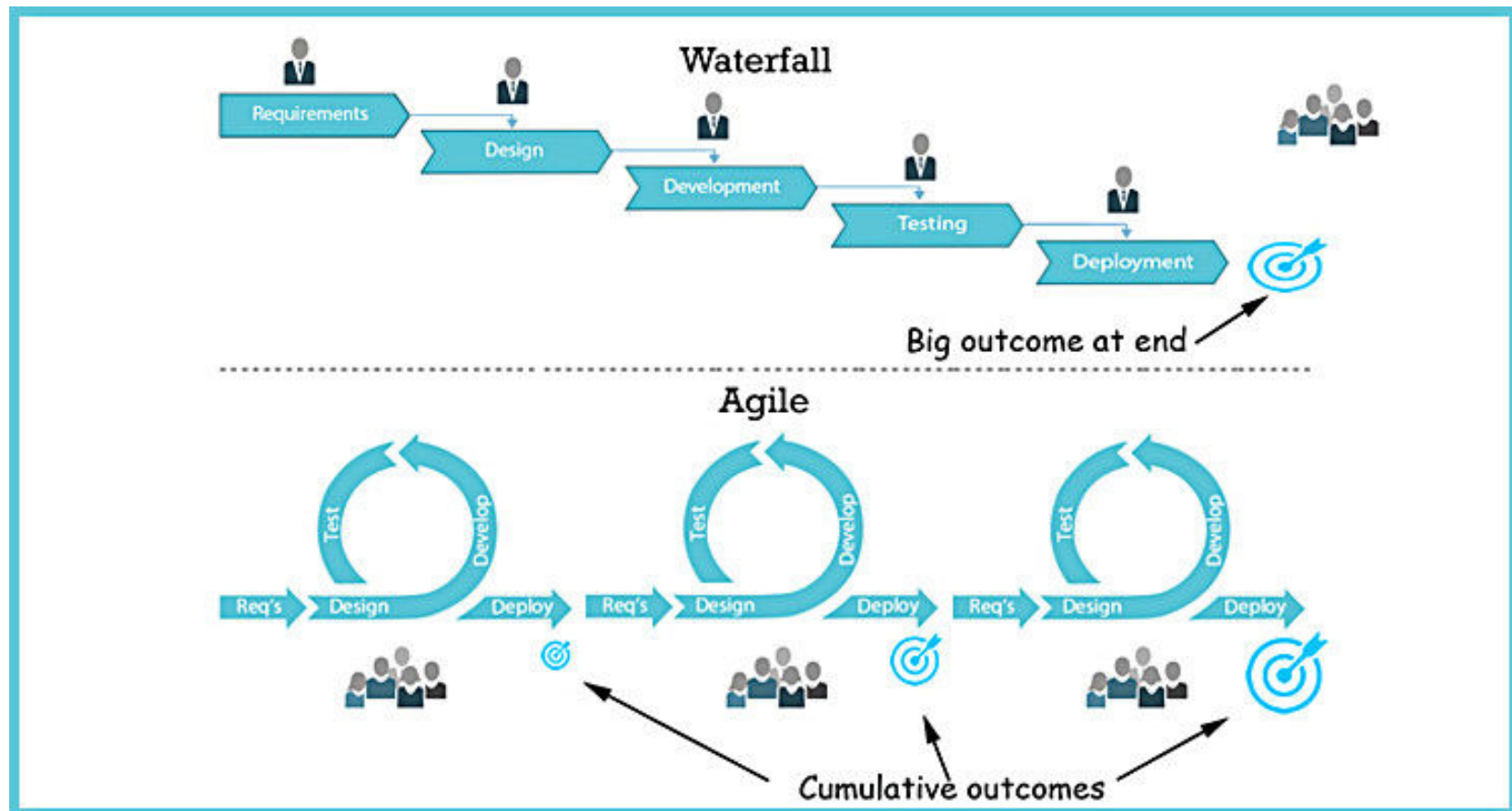
Project lifecycle

Project Lifecycle

- The **series of phases** that a project passes through from its initiation to its closure
- There are many different types of project life cycles, depending on the **industry** , or on the **organization's preferences**.
- ***IT Project:** Requirement -> Design -> Program -> Test -> Implement*
- ***Construction:** Feasibility-> Planning -> Design -> Production -> Turnover -> Startup*



Waterfall vs Agile

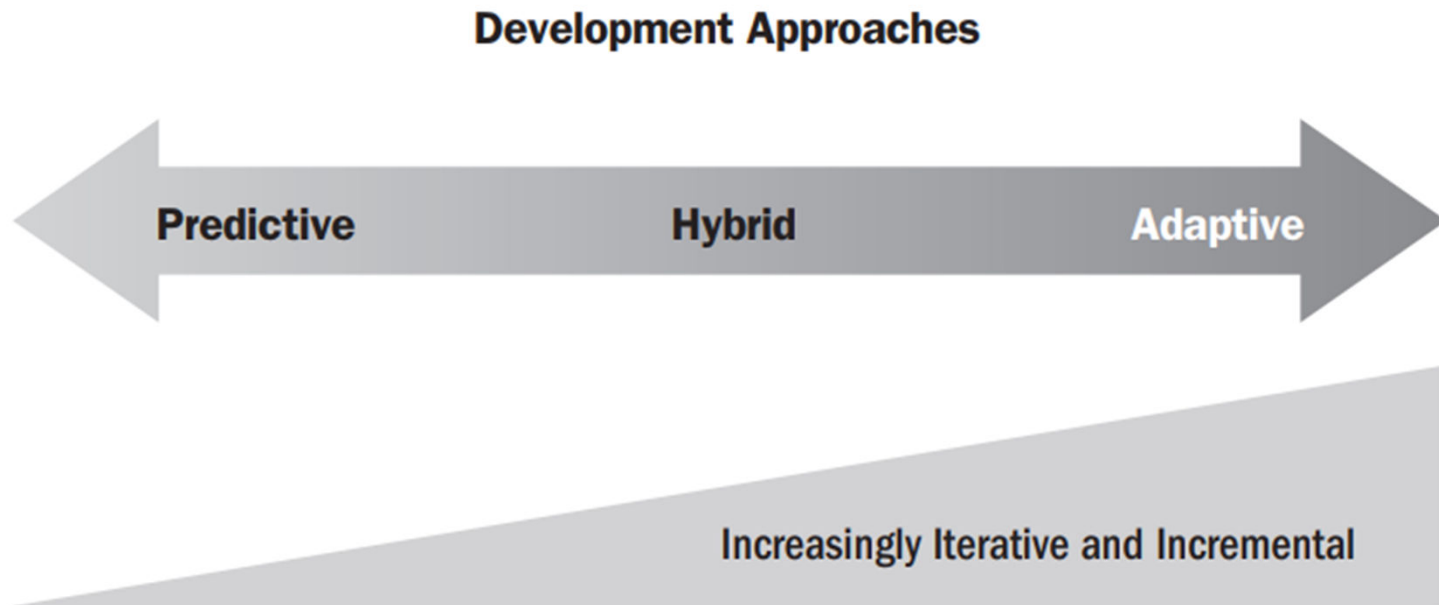


Development approaches

- A development approach is the means used to **create and evolve** the product, service, or result during the project life cycle.
 - Predictive
 - Adaptive
 - Hybrid
- The **goal of project management** is to produce business value in the best possible way given the current environment. It does not matter if that way is agile or predictive. The question to ask is: “**How can we be most successful?**”

Predictive vs Adaptive

Predictive	Agile/adaptive
Process-based	Principle-based
Definable work	High-uncertainty work
Plan-driven	Value-driven
Prescriptive	Descriptive

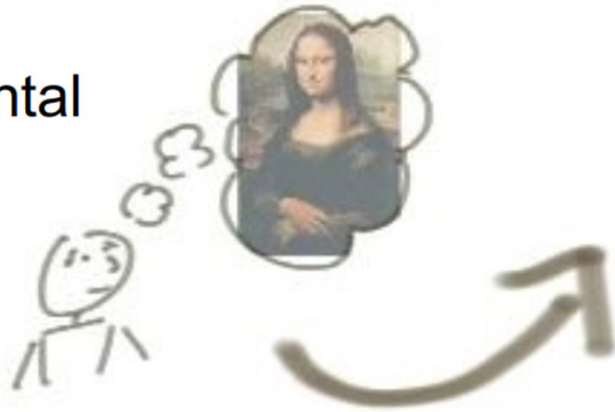


Hybrid approach

- Combination of adaptive and predictive approaches.
 - There is uncertainty or risk around the requirements.
 - Deliverables can be modularized
 - Deliverables that can be developed by different project teams.
- Often use an iterative or / and incremental development approach.
 - **Iterative approach:** clarifying requirements and investigating various options.
 - **Incremental approach:** producing a deliverable throughout a series of iterations.

Incremental & Iterative

Incremental



1



2



3



Iterative



1



2



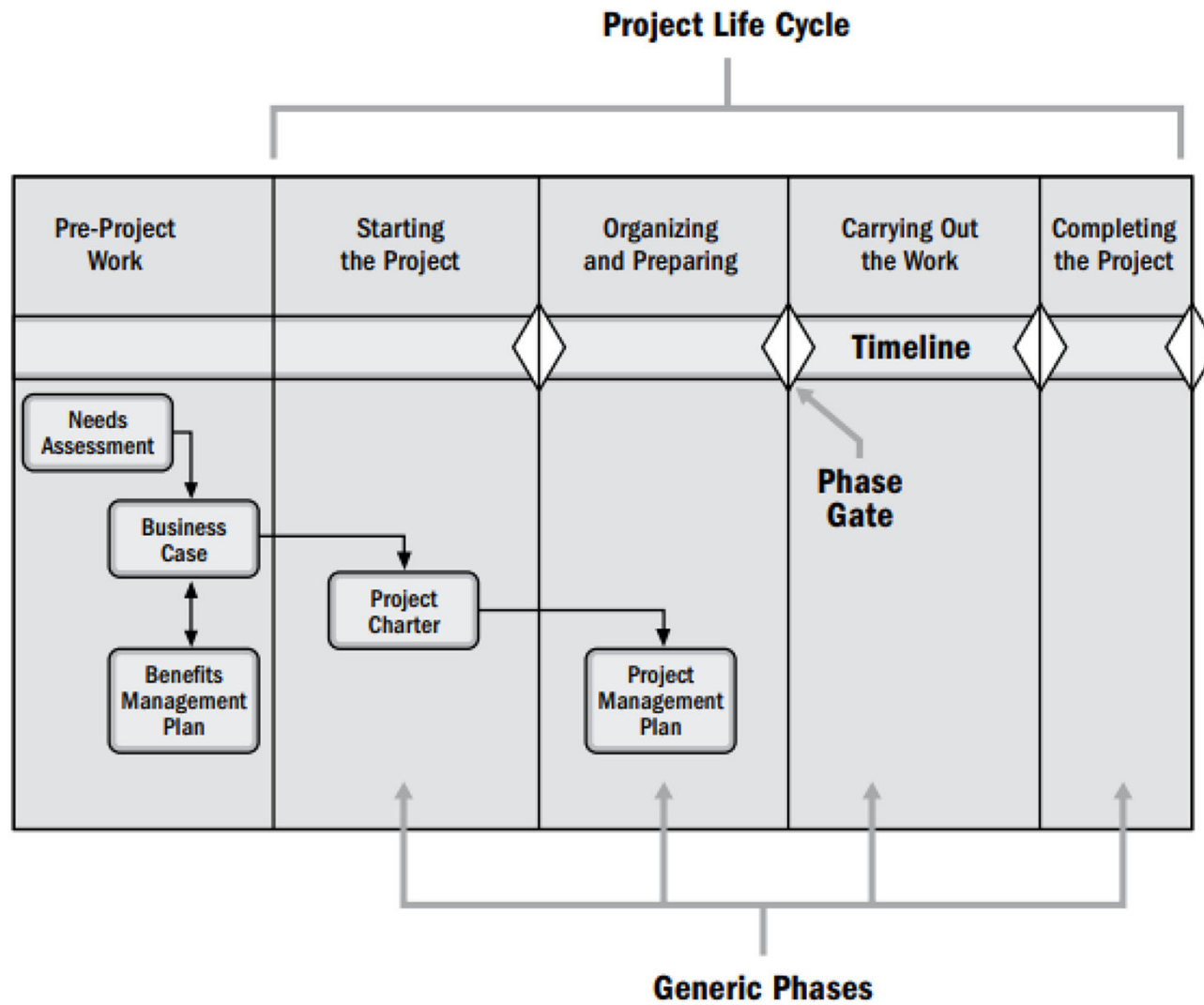
3



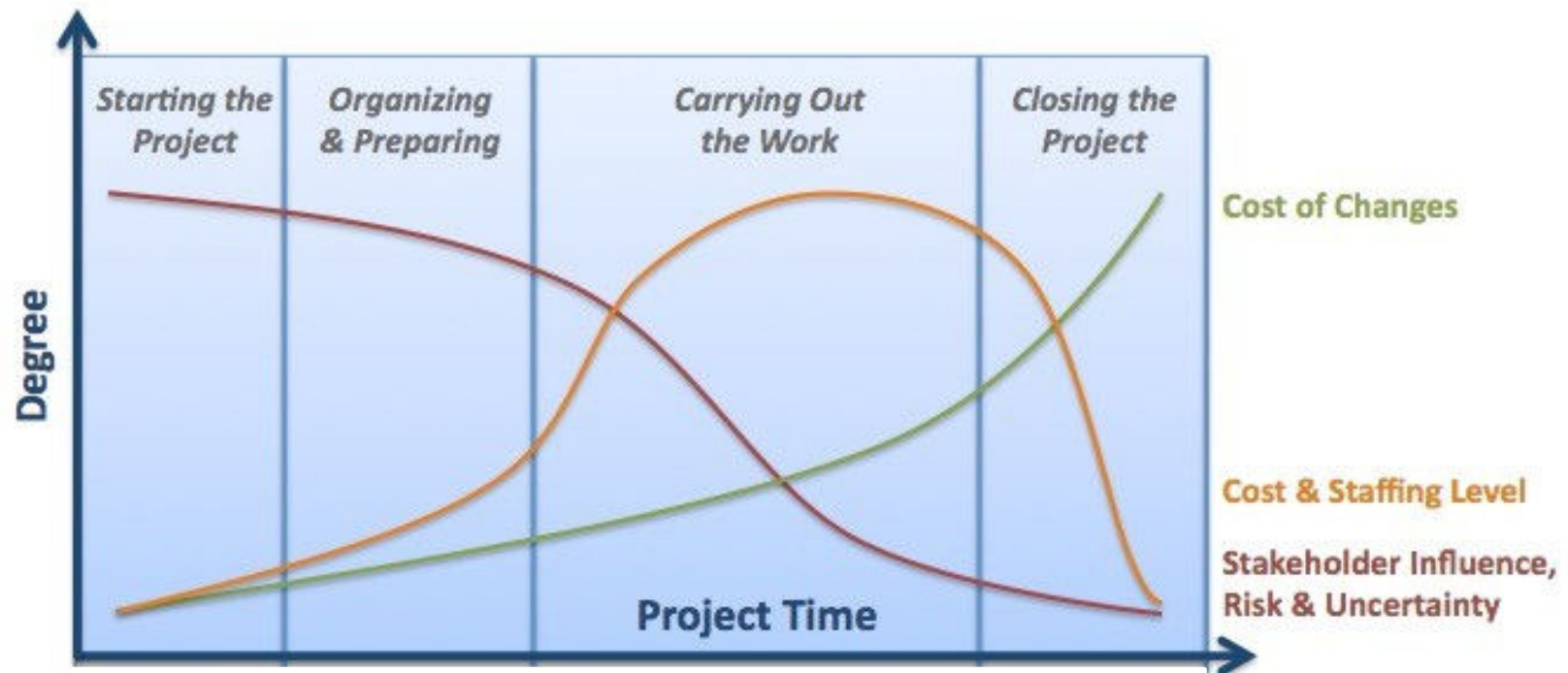
Select a development approach

- Product, service, or result
 - Requirements certainty
 - Ease of change
 - Risk and regulation...
- Project
 - Stakeholders involvement
 - Schedule, funding constraint
- Organization
 - Organizational structure and culture.
 - Organizational capability: suitable policies, ways of working, reporting structure, and attitude
 - Project team size and location.

Project Lifecycle



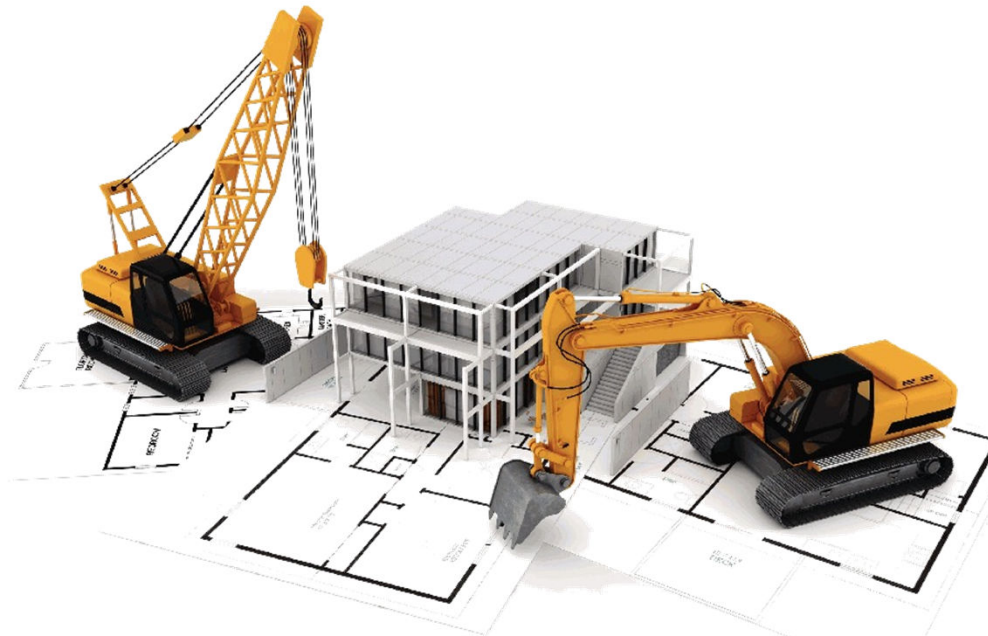
Characteristics of Project Lifecycle



Generic model of project lifecycle

Project phase

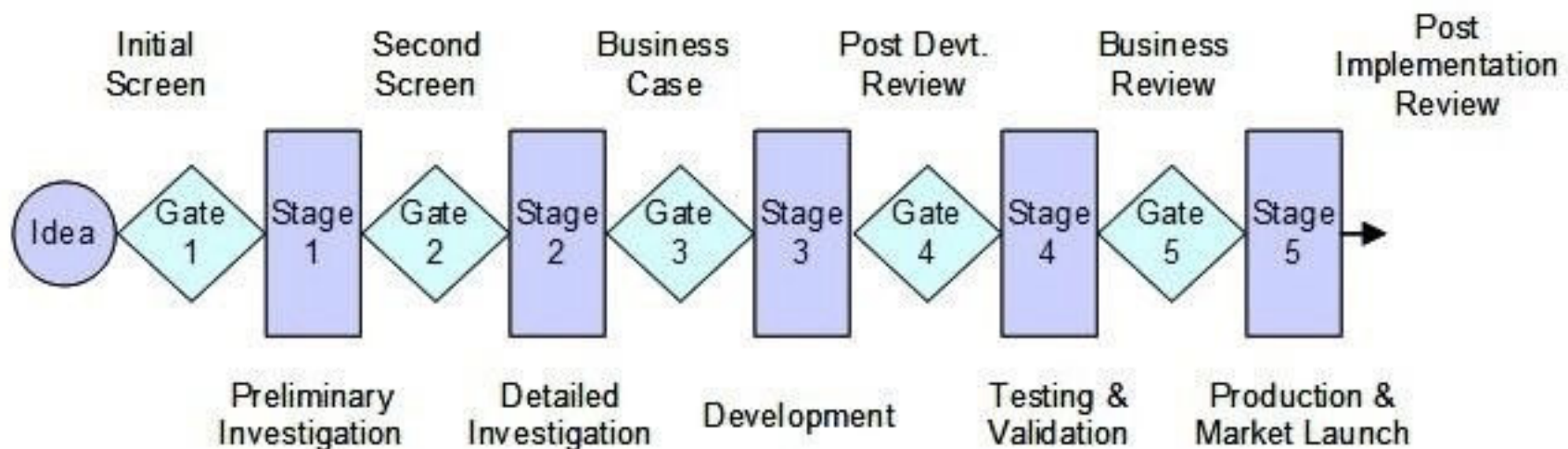
- A collection of logically related project activities that culminates in the completion of **one or more deliverables**
- The transition from one phase to another within a project's lifecycle generally involves some form of technical transfer or handoff. These are also called as **phase gate**



Construction: Feasibility-> Planning -> Design -> Production -> Turnover -> Maintenance

Phase gate

- A review **at the end of a phase** in which a decision is made to continue to the next phase, to continue with modification, or to end a project or program.
- Maybe referred to by other terms: Phase review, stage gate, kill point, phase entrance or phase exit.
- Decision made maybe: **Go, Hold, Kill, Recycle**



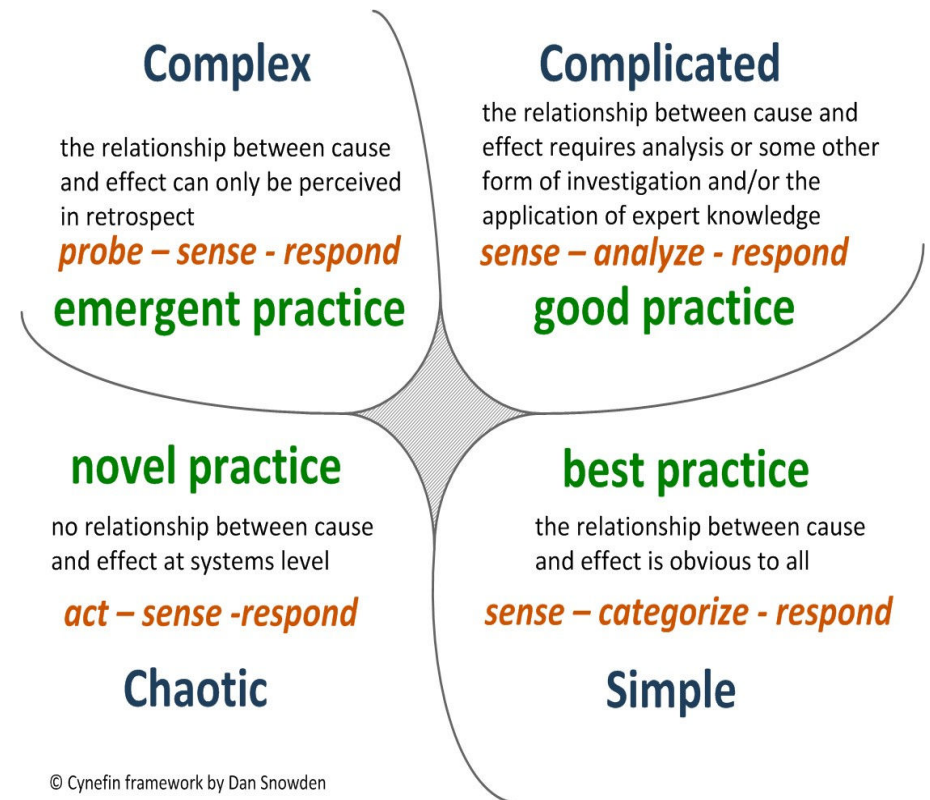
Decision-making: Cynefin framework

Understand the **nature of the context or situation** and choose the most appropriate way to act or respond.

- **Clear (Simple):** Known cause-and-effect; stable; best practices exist
- **Complicated:** Knowable but requires analysis or expertise
- **Complex:** Cause-and-effect only clear in hindsight; emergent behavior
- **Chaotic:** No clear cause-and-effect; urgent crisis
Confused / Disorder: Unclear which domain applies

innovative or strategic initiatives

traditional projects



crisis-management mode

well-understood and highly predictable

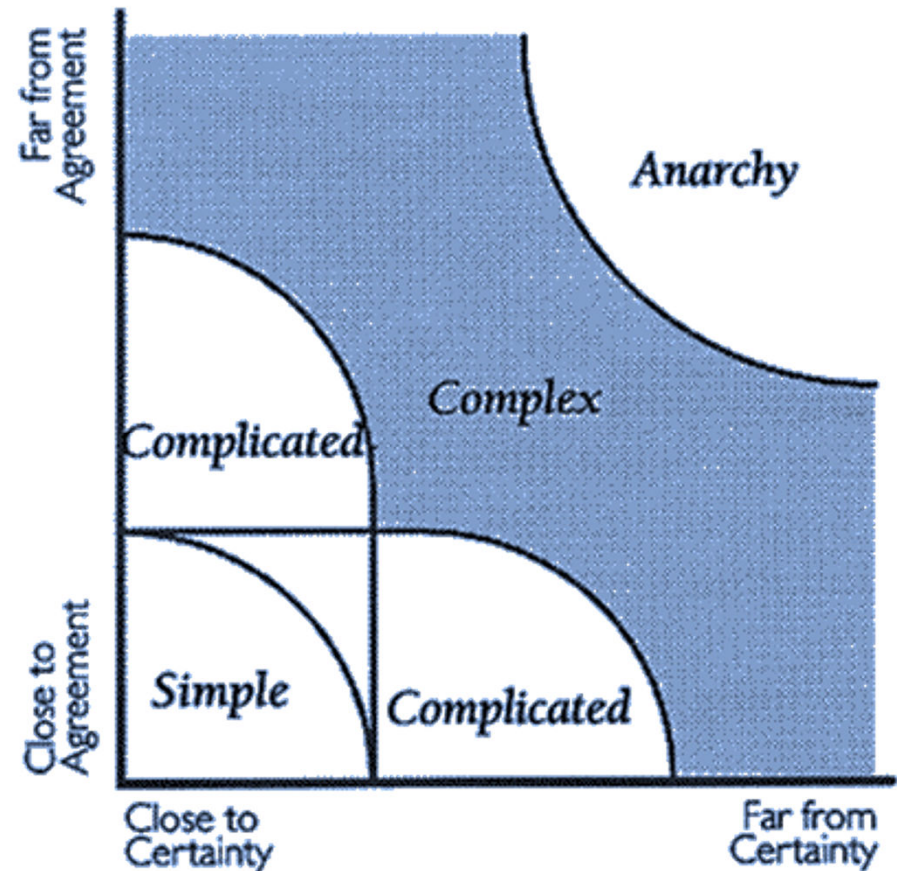
Cynefin framework

Phase	Application of Cynefin	Decision Strategy
Initiation / Gate 0	Identify the domain: Are requirements obvious (clear ROI)? Or are they complex (emerging tech)?	Obvious → Standard project templates Complex → Probe & prototype before committing
Planning / Gate 1	Analyze problem complexity: Are stakeholder needs aligned or evolving?	Complicated → Bring in experts Complex → Run experiments, co-design sessions
Design / Gate 2	Monitor for complexity creep: Are unexpected behaviors emerging?	Adjust project plan, allow iteration and feedback loops
Implementation / Gate 3	Sense and respond: Are solutions producing unpredictable results?	Stay adaptive. For complex areas, allow continuous discovery
Testing & Closure / Gate 4	Determine if results are aligned with original or evolving goals	Accept emergence in complex systems. Focus on learning rather than control

Decision-making: Stacey Complexity Matrix

Analyze the level of complexity and uncertainty to **choose the most appropriate** management or leadership style based on how well-defined the goals and solutions are.

- **Simple:** cause and effect linkages can be clear
- **Complicated:** cause and effect requires analysis or expertise; there is a range of right answers.
- **Complex:** Cause and effect can only be deduced in retrospect. There are no right answers.
- **Anarchy:** Cause and effect are unclear



Decision-making: Stacey Complexity Matrix

Area on Matrix	Implication at Phase Gate	Decision Strategy
Simple (High agreement, High certainty)	Well-known product upgrades, compliance projects	Use standard processes , fast tracking allowed
Complicated (High agreement, Low certainty)	Known goals but technically complex	Use experts , phased prototyping, heavier planning
Complex (Low agreement, Low certainty)	Innovation, R&D, shifting markets	Iterative, agile methods . Keep decision gates flexible
Chaotic (Low agreement, High uncertainty)	Crisis response, urgent launches	Command-based decisions initially, stabilize then move to complex approach

Integration into the Phase Gate Process

- **Data Collection:** Gather progress, cost, risk, and stakeholder feedback.
- **Mapping:** Identify the project's domain in the Cynefin framework or its position on the Stacey Matrix.
- **Impact Analysis:** Link findings to the organization's risk appetite and strategy.
- **Gate Decision:** Choose Go / Hold / Recycle / Kill based on the selected framework.
- **Post-Gate Actions:** Execute the decision and record lessons learned.

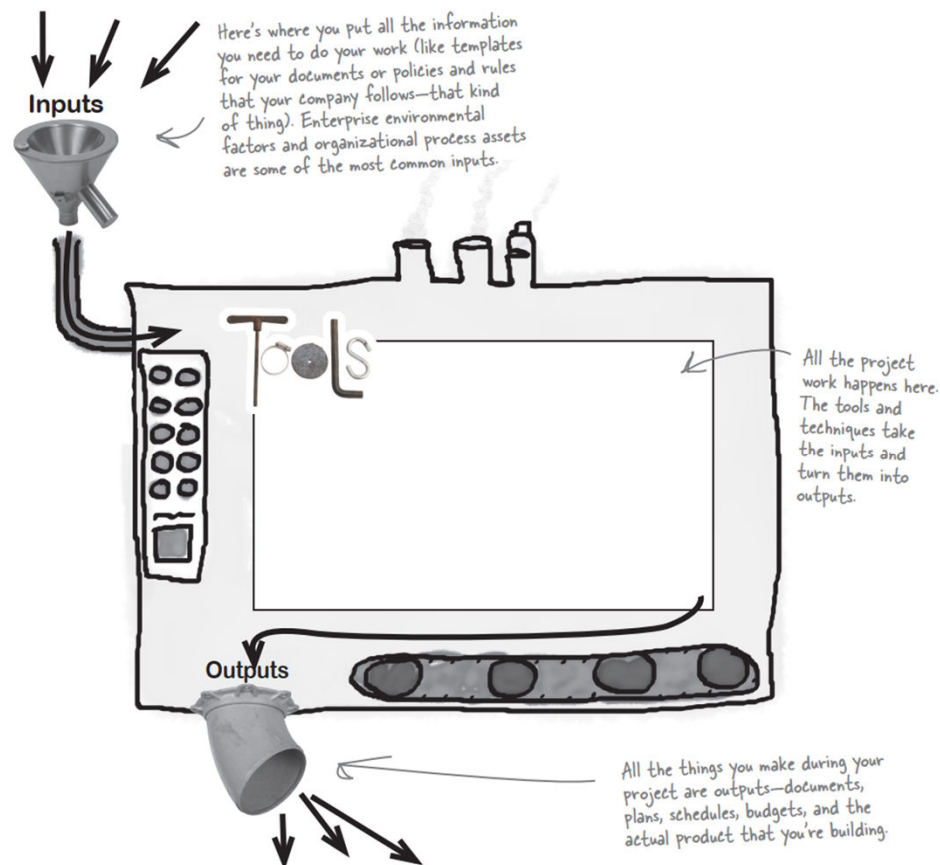
Summary

- Project lifecycle
 - ⇒ *Requirement -> Design -> Program -> Test -> Implement*
- Typical development approaches
 - ⇒ *Predictive & Adaptive (Incremental & Iterative)*
- Phasagate
 - ⇒ *Decision point*

5 Process Groups

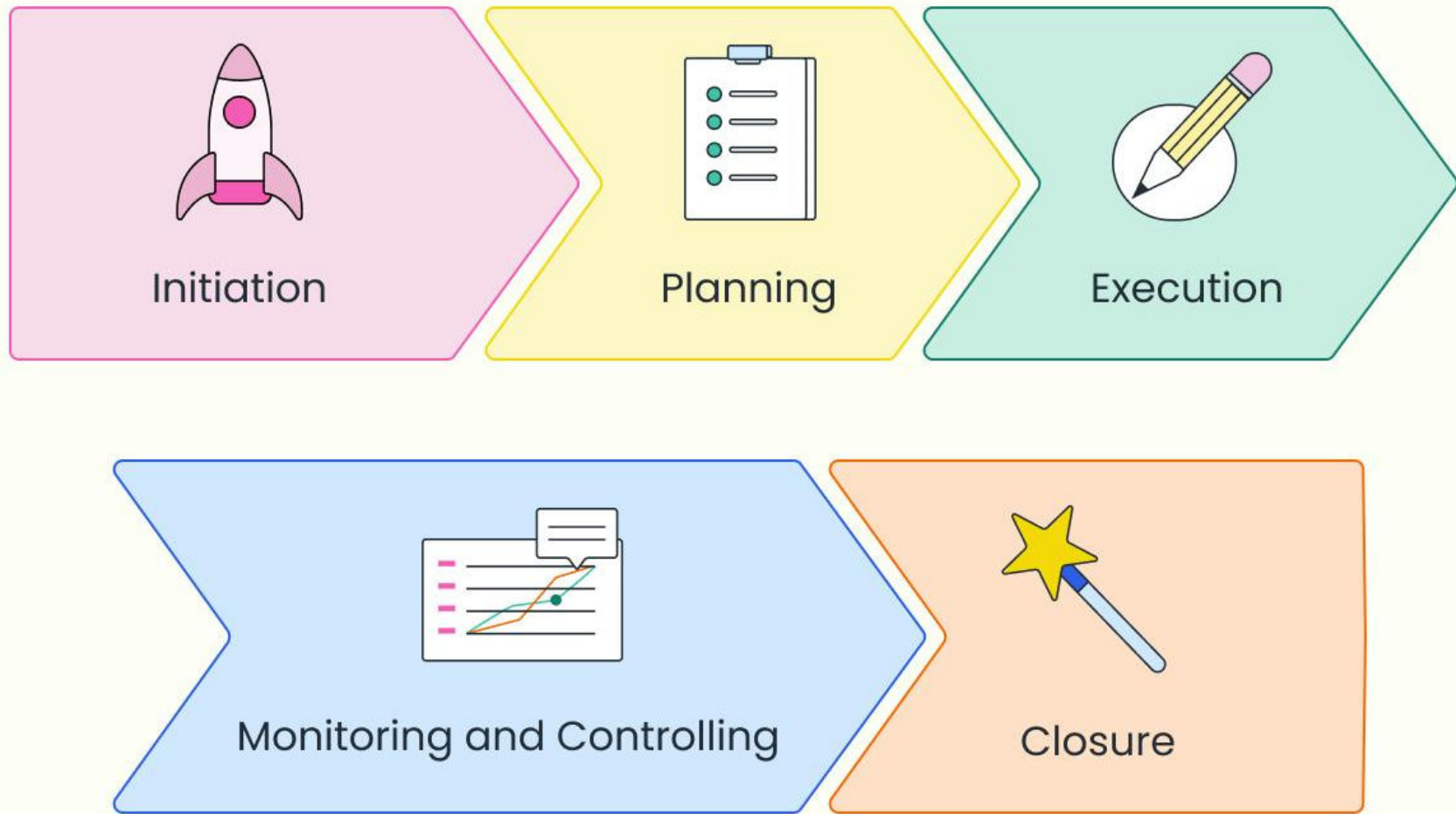
What is a process?

- A **set of** interrelated **actions & activities** performed to create a **pre-specified** product, service or result

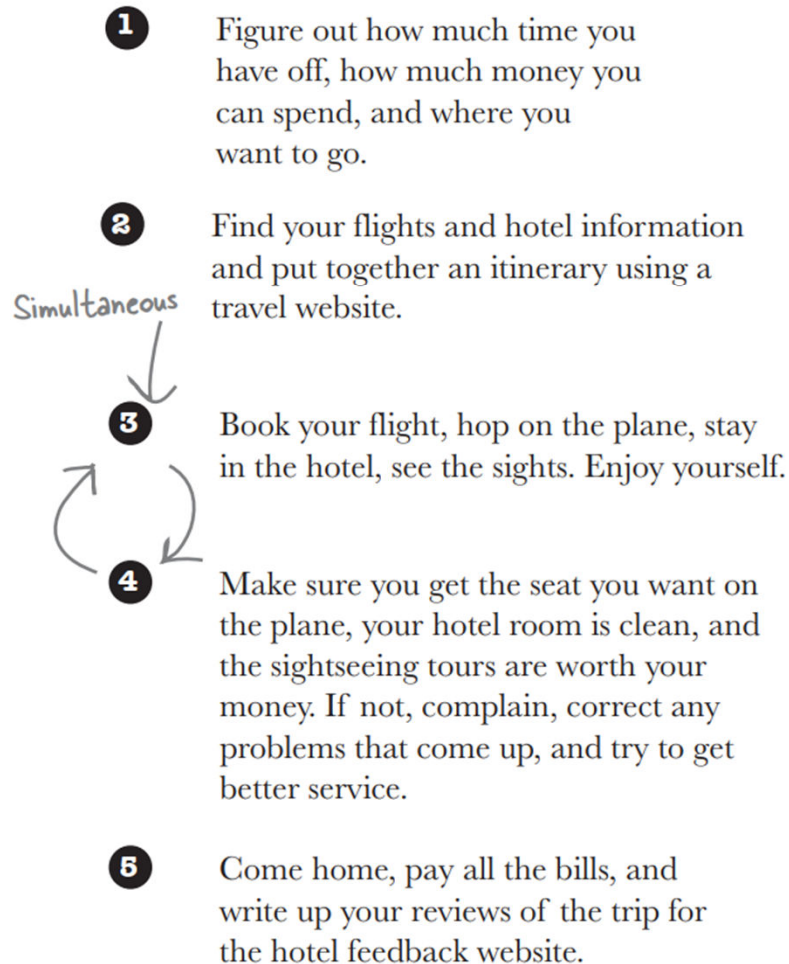


Aspect	Project	Process
Nature	Temporary, unique effort.	Ongoing or recurring activity.
Purpose	To achieve a specific goal or deliverable (often one-time).	To produce consistent results or support ongoing operations.
Start & End	Has a defined beginning and end.	Each execution has clear inputs and outputs, but the process as a system continues as long as needed.
Repetition	Not repeated (unless similar projects are initiated separately).	Can be repeated whenever the triggering condition occurs.
Output	A unique product, service, or result.	Standardized, repeatable outputs.
Flexibility	High — adapts to project-specific needs.	Low to moderate — usually standardized to ensure consistency.
Example	Building a hospital.	Processing a patient admission.

5 Process Groups

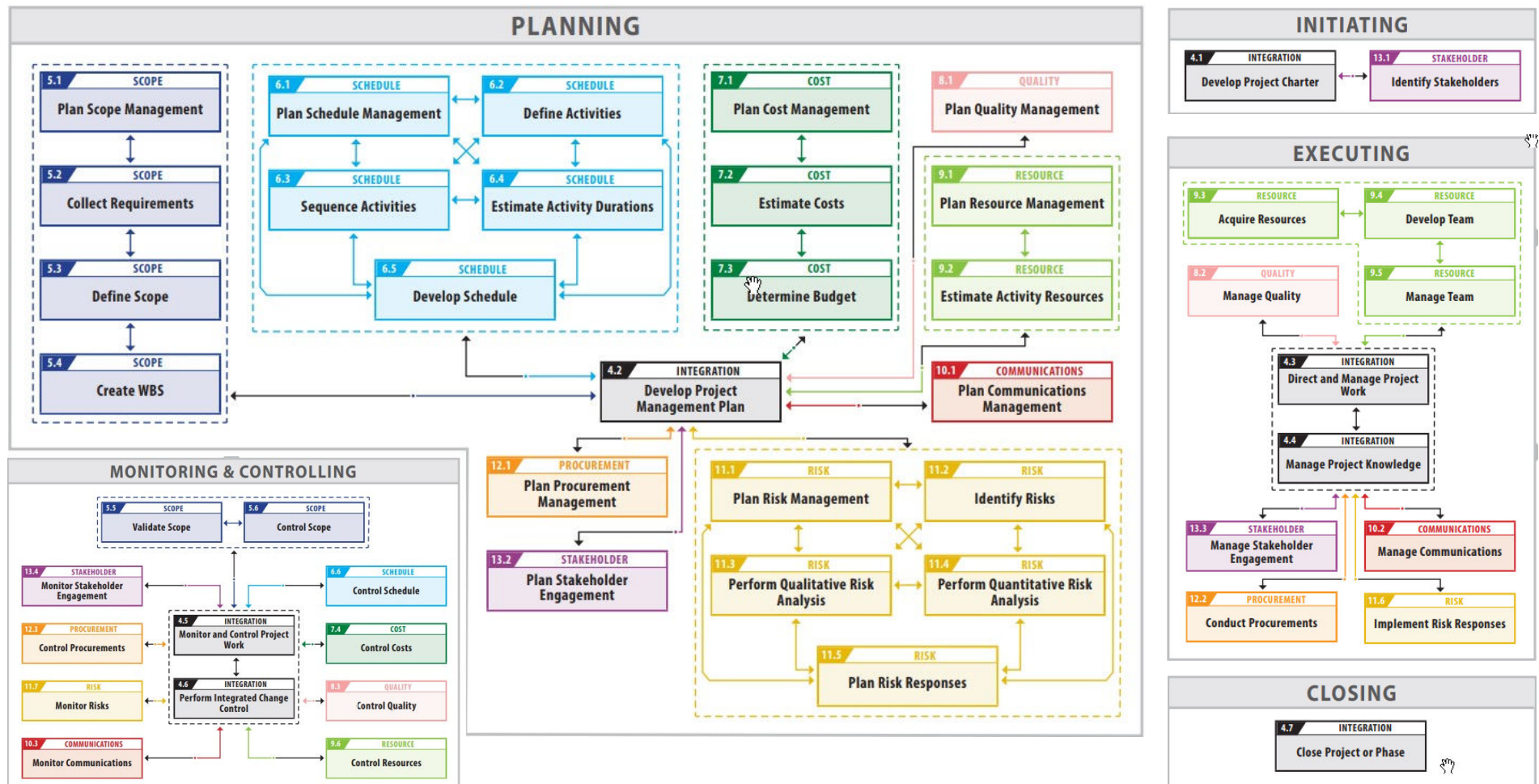


5 Process Groups



- 1** Develop Project Charter.
- 2** Develop Project Management Plan.
- 3** Direct and Manage Project Work.
- 4** Monitor and Control Project Work.
- 5** Close Project.

49 Processes of PMI





Initiating
process group



Planning
process group



Executing
process group



Monitoring
& Controlling
process group



Closing
process group

Develop
Project
Management
Plan

Direct and
Manage Project
Work

Plan Quality
Management

Develop
Project
Charter

Control
Schedule

Monitor
and Control
Project Work

Estimate
Activity
Durations

Control
Scope

Manage
Communications

Identify
Stakeholders

Identify Risks

Close Project
or Phase



Initiating
process group



Planning
process group



Executing
process group



Monitoring
& Controlling
process group



Closing
process group

Develop
Project
Charter

Identify
Stakeholders

There are only
two processes
in the entire
Initiating process
group!

Develop
Project
Management
Plan

Estimate
Activity
Durations

Identify Risks

Plan Quality
Management

Direct and
Manage
Project Work

Manage
Communications

This one was a little
tricky, but if you keep in
mind that the Executing
process is where you do
your work, it makes
more sense.

Control
Scope

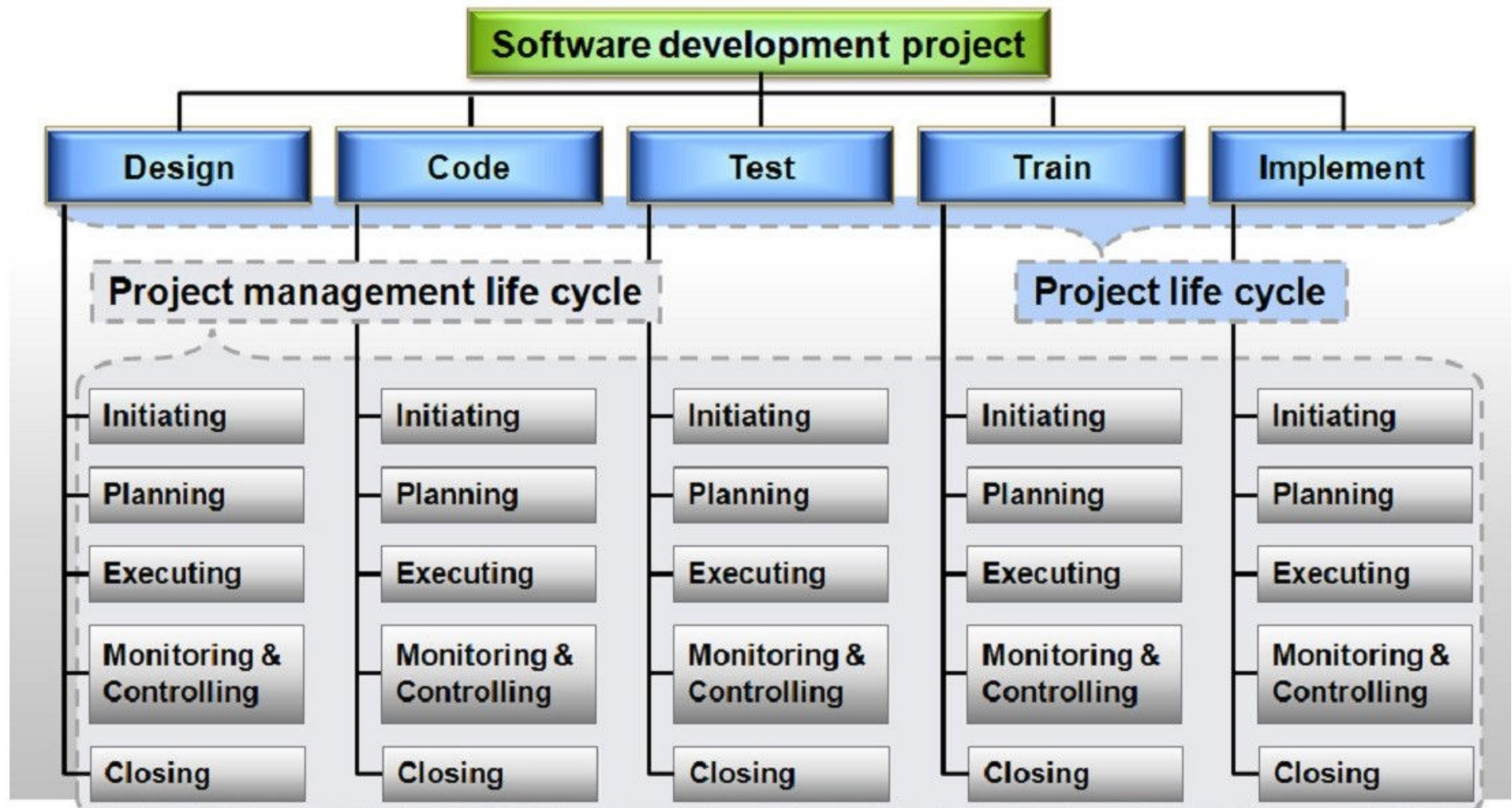
Monitor
and Control
Project Work

Control
Schedule

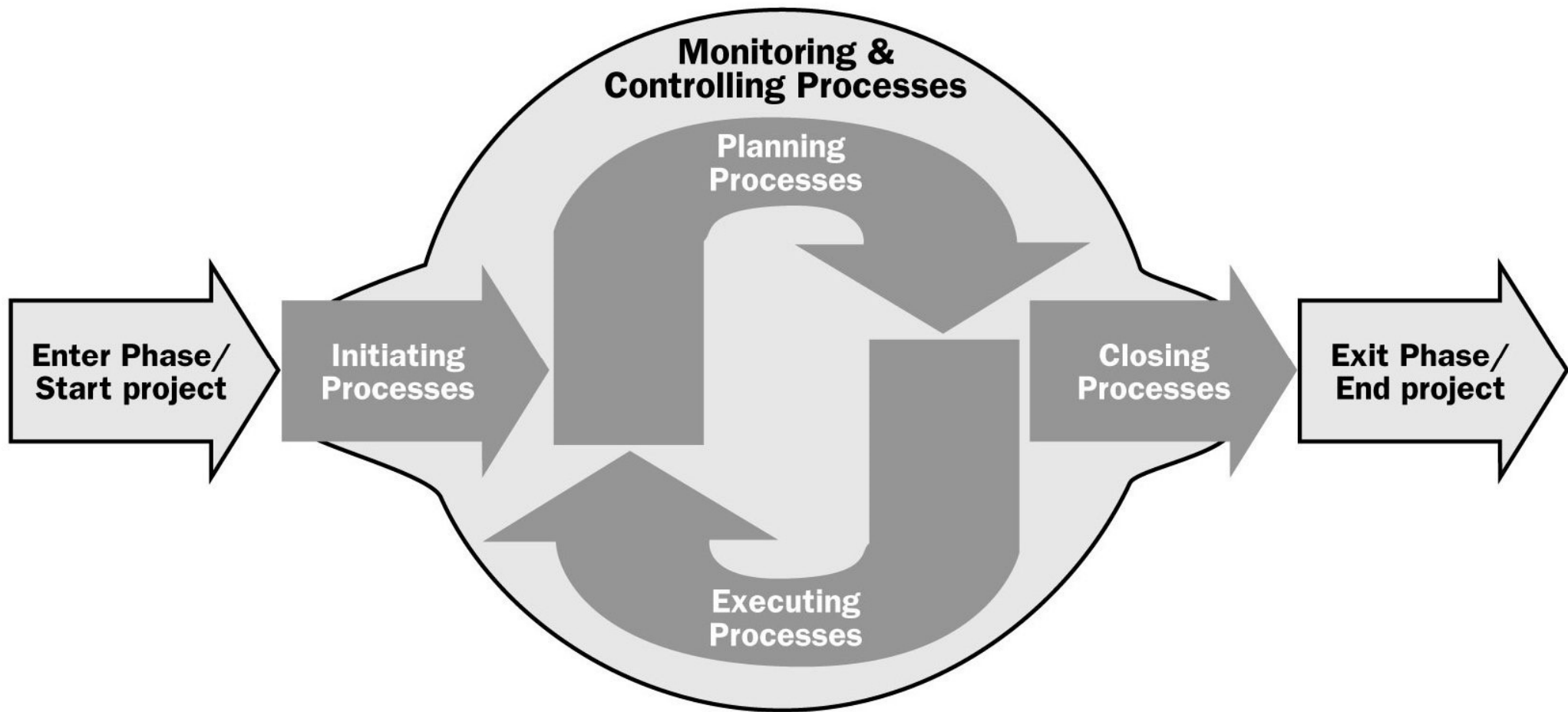
When a process
starts with
"control," it's
part of the
Monitoring and
Controlling
group.

Close
Project
or
Phase

Project Lifecycle vs Project Management Processes



Project Management Processes Interactions



Process Group interactions Within a Project or Phase

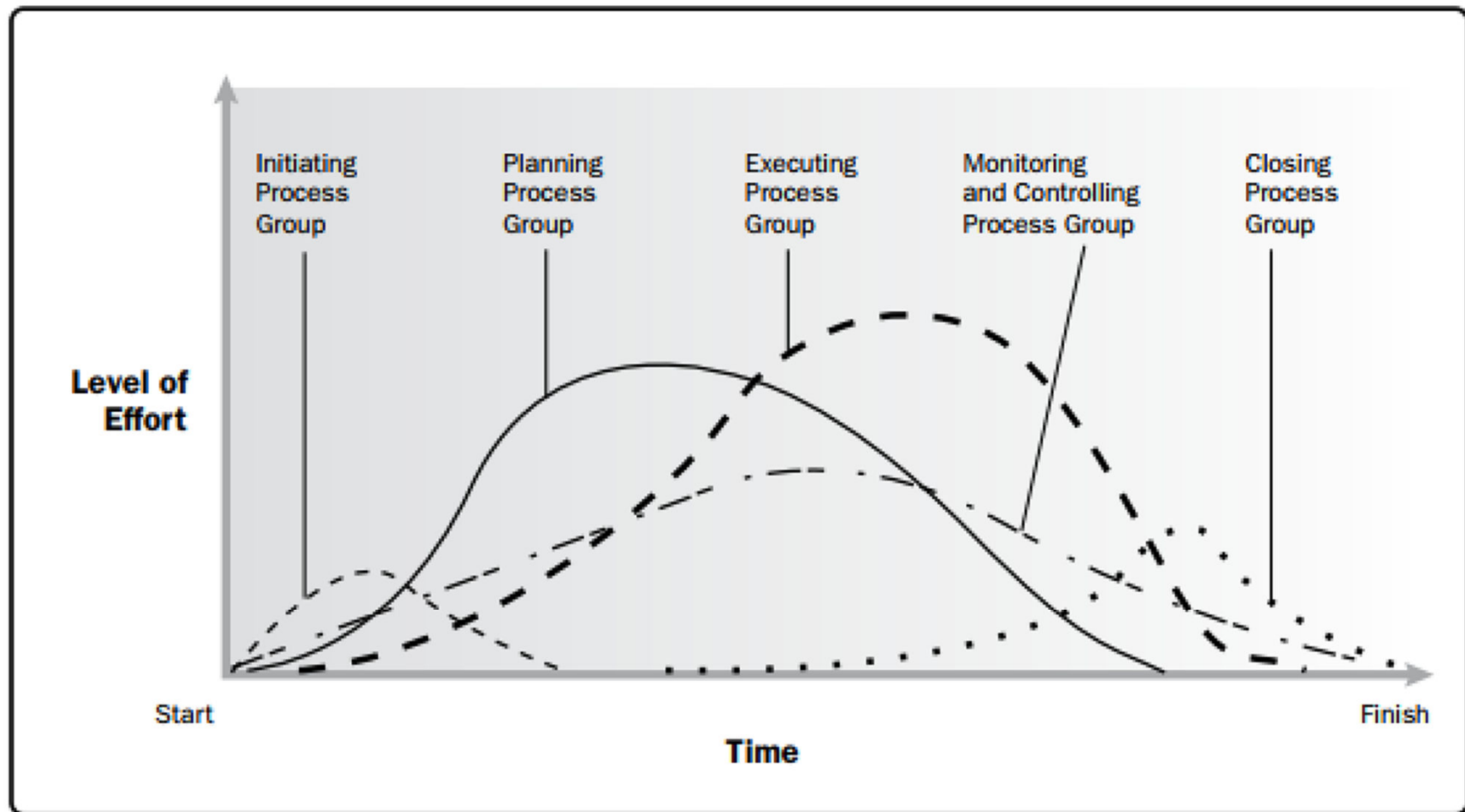
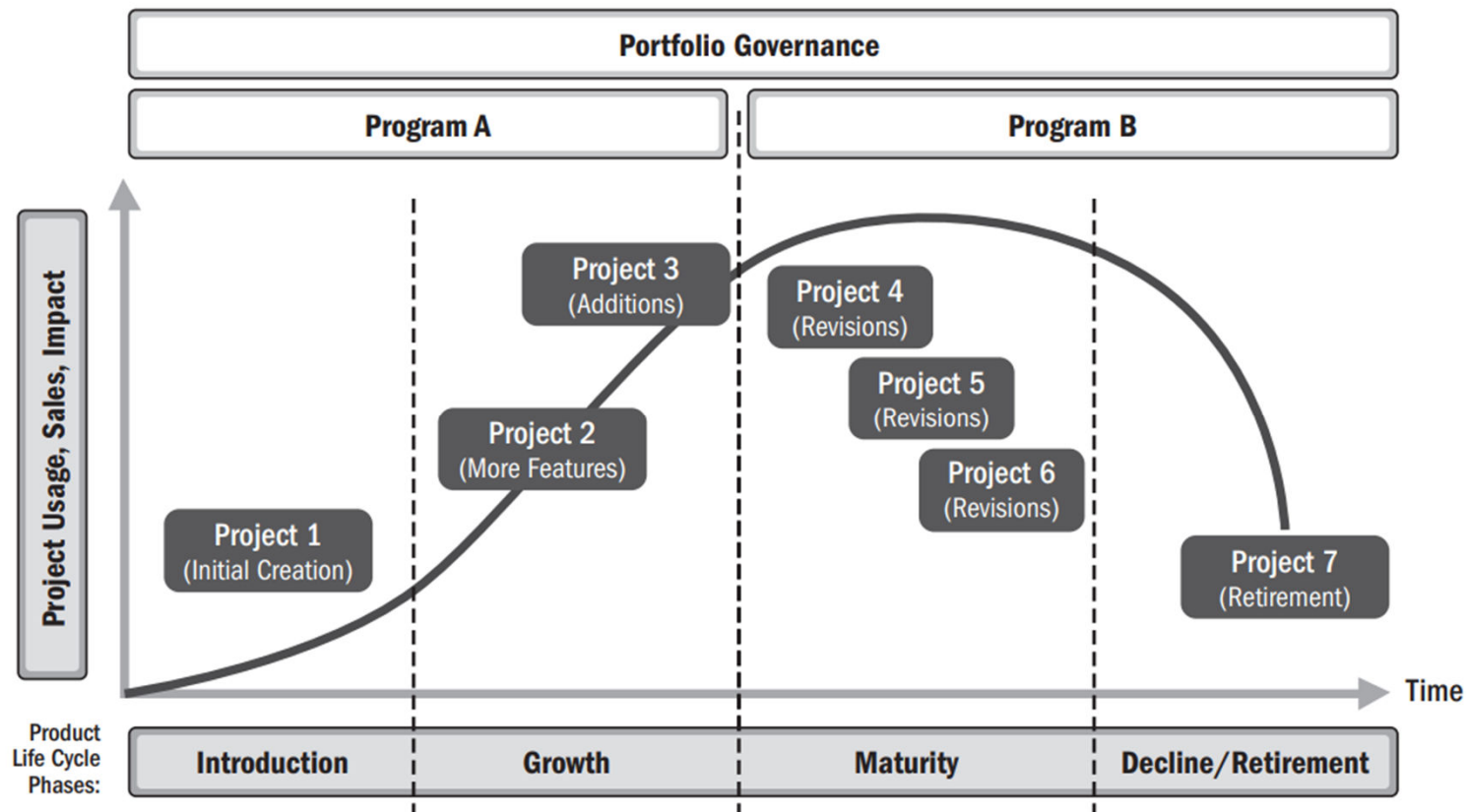


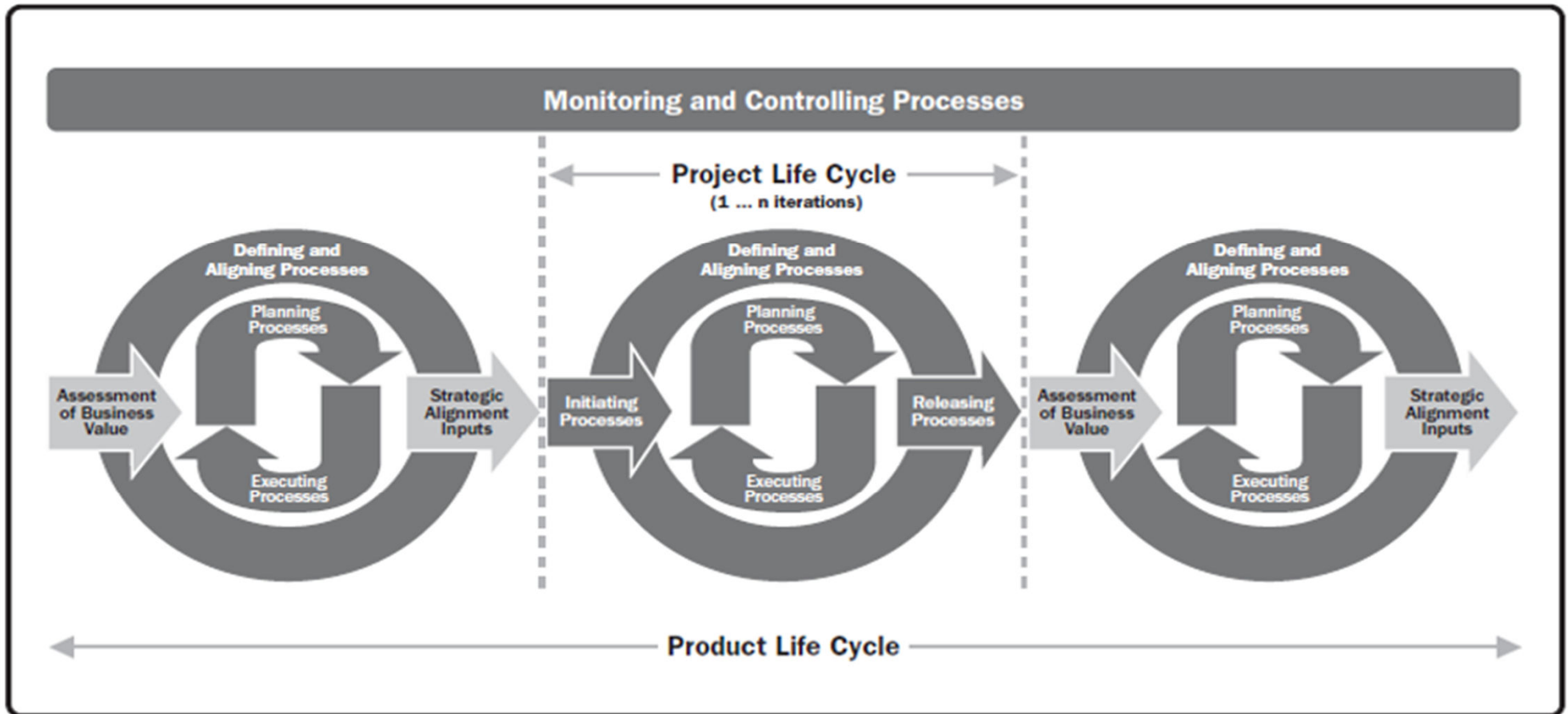
Figure 1-5. Example of Process Group Interactions Within a Project or Phase

Product Lifecycle

- Product lifecycle is the series of phases that represent the evolution of a product, from concept through delivery, growth, maturity and to retirement



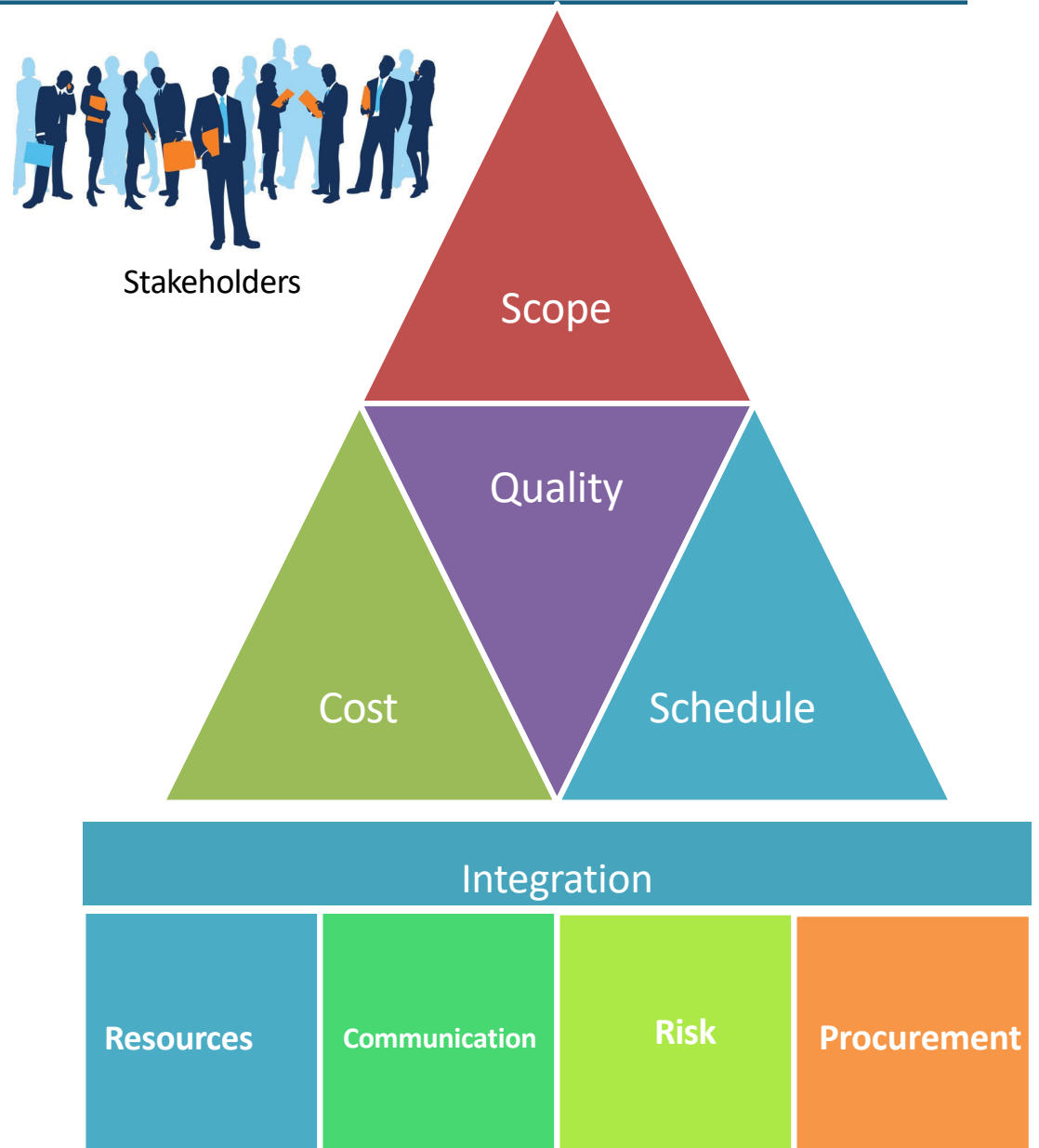
Product and Project lifecycle



Project Managment Knowledge Areas

Project Management Knowledge Areas

- The **process groups** help you organize the processes by the *kind of work* you do.
- The **knowledge areas** help you organize by the *subject matter* you're dealing with.



Cookie Making Project

Integration

Making sure all the right parts of the project come together in the right order, at the right time

Scope

Everything that's included in the project, and nothing that's excluded.

Cost

Budgeting for the cookie project

Project Resource

Making sure your schedule is clear, and your oven is available for cooking

Risk

Could you burn the cookies or yourself on the range? Are the eggs fresh?

Stakeholder

Does your sweetheart like chocolate chip cookies, peanut butter, or oatmeal?

Procurement

Selecting the right store to supply your ingredients

Schedule

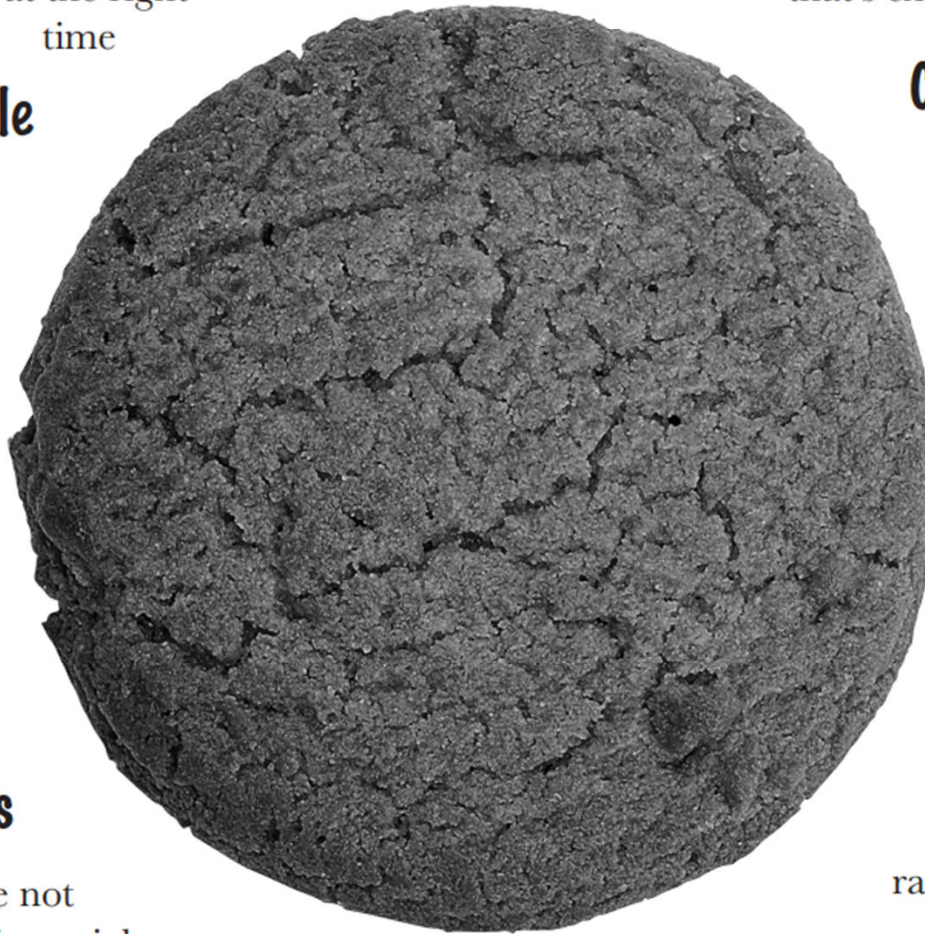
Preparation and cooking time

Quality

Checking that the cookies look and taste right

Communications

Making sure you're not mixing metric and imperial measurements



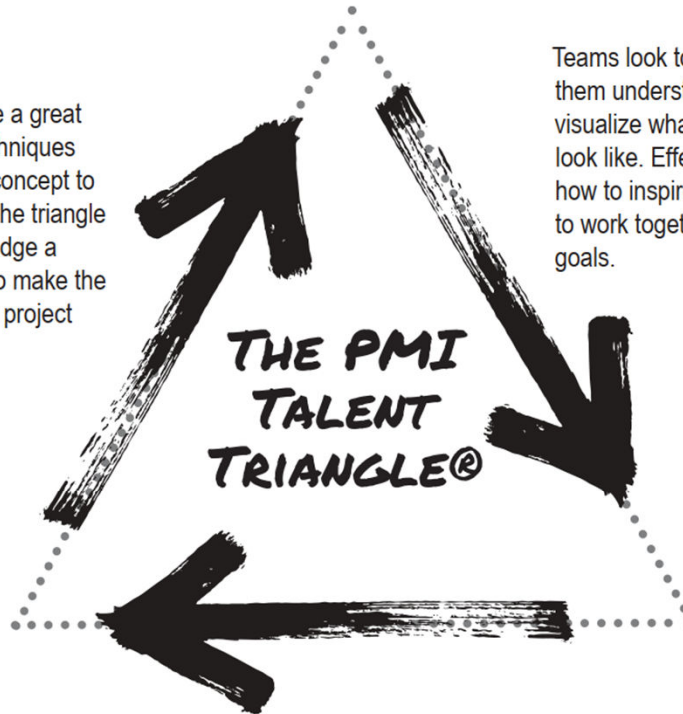
PMI Talent Triangle

Technical Project Management

Effective project managers have a great grasp on all of the tools and techniques that guide a project from initial concept to finished project. This aspect of the triangle is all about the technical knowledge a project manager needs to use to make the right decisions when directing a project team's work.

Leadership

Teams look to project managers to help them understand the project's goals and visualize what a successful outcome will look like. Effective project managers know how to inspire the teams they work with to work together to accomplish strategic goals.



Strategic and Business Management

Keeping a project on track means understanding the strategic purpose behind it. Successful project managers know how to interpret the environment of the organizations they work in and how to navigate that organization and position their projects for success. They constantly keep tabs on the project's direction and make sure that it's delivering as much value as possible for the organization that's sponsoring it.

Styles of leadership

Laissez faire

These leaders are “hands-off,” leaving most of the goal setting and daily leadership of the team to the team itself. They step in and make decisions only as a last resort.



Transactional

Leaders who are transactional make goals explicit and then track progress toward completing them. When the team's actual progress deviates from the leader's plan, he or she will provide course correction.

Charismatic

Charismatic leaders are inspiring because they hold strong convictions and have magnetic personalities. They use that inspiration to create a team culture that encourages innovation and aligns the team with those strongly held convictions and goals.



Transformational

Like charismatic leaders, transformational leaders focus on inspiring their teams with idealized goals. But transformational leaders encourage their teams to change the environment around them. Transformational leaders work to motivate each team member to be creative and make change happen.

Interactional

This type of leadership changes depending on the situation. Interactional leaders take whatever approach makes sense in the given situation.



Servant leader

This type of leadership is most often discussed in relation to agile thinking and methods. Here, a leader focuses on serving the people on the team. By clearing roadblocks and helping team members get what they need to achieve the group's goal, the servant leader promotes better relationships on the team and creates a collaborative team culture.

Summary

- 5 Process Groups
- 10 Project Management Knowledge Areas
- PMI talent triangle

Summary

- 5 Process Groups
- 10 Project Management Knowledge Areas
- PMI talent triangle

